

Timing diagram for the 4096-word memory access. The diagram shows four signals: CLK (clock), q4 (address bit 4), q (16-bit address bus), b (data bus), and a (16-bit data bus). The address bus q is shown as a sequence of 16-bit values from 0 to 15, repeating twice. The data bus b is shown as a sequence of 16-bit values from 0 to 1, repeating twice. The data bus a is shown as a sequence of 16-bit values from 0 to 14, repeating twice. The diagram is divided into three sections: 0 - 30, 0 - 3, and DEC.

Timing diagram for the 4096-word memory access sequence. The diagram shows four signals: CLK (clock), q4 (bus 4), q (bus), b (bus), and a (address). The address bus 'a' is shown as a sequence of 31 values: 0-15, 0, 1-14. The data bus 'q' shows 31 data words, each a 16-bit value. The bus 'b' shows a single bit value, 0 for the first 16 cycles and 1 for the next 16 cycles. The clock 'CLK' is a periodic square wave. The diagram is divided into three sections: 0-30, 0-3, and DEC.

Timing diagram for the 4096-word memory access sequence. The diagram shows four signals: CLK (clock), q4 (bus 4), q (bus), b (byte enable), and a (address). The address 'a' is shown as a sequence of 32 values (0-15, 0-14) over 32 clock cycles. The bus 'q' shows the data returned for each address. The byte enable 'b' is 0 for the first 16 cycles and 1 for the next 16 cycles. The clock 'CLK' is a periodic square wave. The diagram is divided into two sections: '0 - 30' and '0 - 3', with a 'DEC' label at the end.

Timing diagram for the 4096-word memory access sequence. The diagram shows four signals: CLK (clock), q4 (bus 4), q (bus), b (bus), and a (address). The address bus 'a' is shown as a sequence of 31 values: 0-15, 0, 1-14. The data bus 'q' shows 31 data words, each 16 bits wide, corresponding to the address sequence. The bus 'b' shows a single bit value, 0 for the first 16 cycles and 1 for the next 15 cycles. The clock 'CLK' is a periodic square wave. The diagram is divided into three time intervals: 0-30, 0-3, and DEC.

Timing diagram for the 4096-word memory access sequence. The diagram shows four signals: CLK (clock), q4 (bus 4), q (bus), b (byte enable), and a (address). The address bus 'a' is divided into two 16-bit segments. The first segment (0-15) is active during the first 16 clock cycles, and the second segment (0-14) is active during the next 15 clock cycles. The bus 'q' shows the data being read from memory, and 'b' shows the byte enable signal, which is 0 for the first 15 cycles and 1 for the last cycle of the first segment.

Timing diagram for the 4096-word memory access sequence. The diagram shows four signals: CLK (clock), q4 (bus 4), q (bus), b (bus), and a (address). The address 'a' is shown as a sequence of 31 values (0-15, 0-14) over 31 clock cycles. The bus 'q' shows the data returned for each address. The bus 'b' shows the data returned for the address 0. The bus 'q4' shows the data returned for the address 4. The diagram is divided into three sections: 0 - 30, 0 - 3, and DEC.

Timing diagram for the 4096-word memory access sequence. The diagram shows four signals: CLK (clock), q4 (bus 4), q (bus), b (bus), and a (address). The address 'a' is shown as a sequence of 31 values (0-15, 0-14) over 31 clock cycles. The bus 'q' shows the data returned for each address. The bus 'b' shows the data returned for the address 0. The bus 'q4' shows the data returned for the address 4. The diagram is divided into three sections: 0 - 30, 0 - 3, and DEC.

Timing diagram for the 4096-word memory access sequence. The diagram shows four signals: CLK (clock), q4 (bus 4), q (bus), b (byte enable), and a (address). The address bus 'a' is divided into two 16-bit segments. The first segment (0-15) is active during the first 16 clock cycles, and the second segment (0-14) is active during the next 15 clock cycles. The bus 'q' shows the data being read from memory, and 'b' shows the byte enable signal, which is 0 for the first 15 cycles and 1 for the last cycle of the first segment.



















